



Redesigning Tea Plucking Bucket

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The Importance of Tea Plantations

UNDERSTANDING THE ROLE OF WOMEN WORKERS

Tea plantation work is a major source of livelihood in Assam and other parts of India. A large portion of this workforce consists of women who manually pluck tea leaves and carry them in bags weighing approximately 10–15 kg. Traditionally, these bags are supported using a neck or head strap, which transfers a significant portion of the load to the cervical spine.



Why It Continues

CULTURAL FACTORS

Traditional carrying methods are deeply rooted in local culture, where they are perceived as normative practices among workers, influencing their daily routines and identities.

LIMITED ALTERNATIVES

A lack of awareness regarding ergonomic designs and modern alternatives contributes to the persistence of outdated practices, leaving workers unaware of potential improvements in their working conditions.

ECONOMIC FACTORS

Many workers rely on the current methods for their livelihoods, making it difficult to justify investing in new systems that may seem expensive or unnecessary at first glance.

NEED FOR REDESIGN

There is a significant need for affordable, practical redesigns that respect cultural norms while providing greater comfort and safety, ensuring workers can carry loads without risking their health.

Problem Statement: Redesigning Carrying System

MINIMIZE NECK LOAD

Our goal is to effectively **reduce the load** on the neck, cervical spine, and shoulders of workers. By redistributing the weight of the tea leaves, we can alleviate the strain on these vulnerable areas, thus enhancing overall well-being and productivity.

SUITABLE FOR UNEVEN TERRAIN OF TEA PLANTATIONS

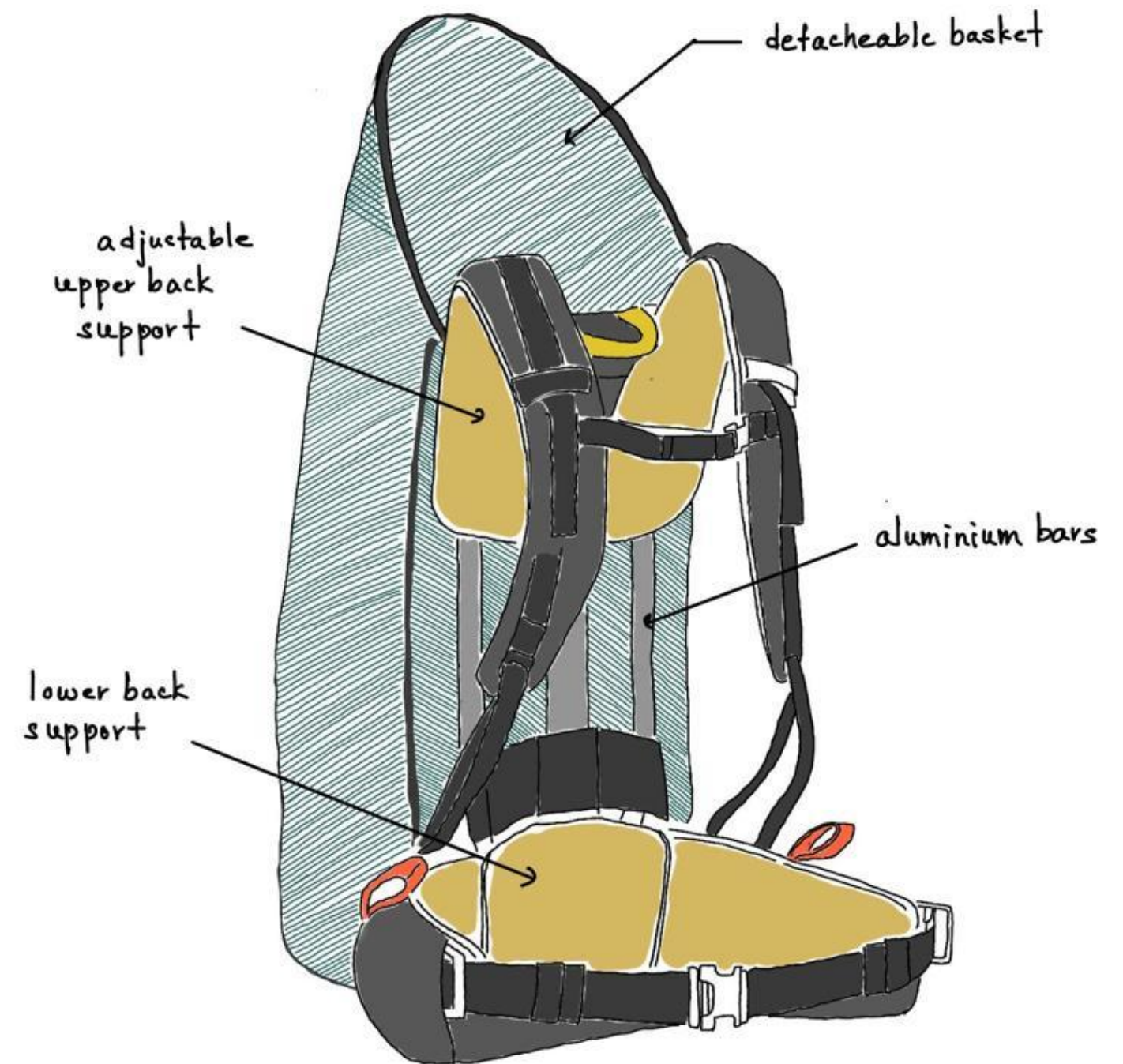
REDISTRIBUTE WEIGHT

We aim to **redistribute the weight** towards stronger body parts, specifically the hips and back. This ergonomic approach ensures that heavy loads are carried more naturally, decreasing the risk of injury and promoting a healthier work posture for tea plantation workers.

Our Design Concept Overview

INTEGRATING ERGONOMICS WITH TRADITIONAL DESIGN

Our redesign retains the **conventional leaf storage** geometry, ensuring seamless workflow compatibility, while enhancing user experience through ergonomic principles for better weight distribution and comfort.



Material & Cost

Bag Body - Polyester Material (Rs 400)

3 Pcs Buckles (Rs 120)

4 Pcs Strap Adjuster (Rs 80)

Aluminum Rod (Rs 150)

Foam Sheet (Rs 60)

Basket (Rs 150)

Total Cost - Rs 960/-



Ergonomic Principles Applied

ALUMINIUM BARS

The inclusion of **aluminium bars** significantly enhances comfort, Provides curved back arc for ergonomic fit Prevents pouch sagging. Transfers load directly to hip belt

DETACHEABLE BASKET

Easily removable for unloading. Retains conventional cylindrical shape (familiar to workers). Mounted securely using straps/hooks. Replaceable if damaged

ADJUSTABLE UPPER BACK SUPPORT

Incorporating **adjustable support** ensures a personalized fit for each user, accommodating different body types and preferences, thereby maximizing comfort and functionality during tea leaf transportation

LOWER BACK SUPPORT

Padded lumbar cushion supports natural spinal curvature. Reduces lower back muscle fatigue. Distributes vertical load to hip belt. Minimizes bending stress on lumbar vertebrae

Benefits of the Redesigned System

REDUCED STRAIN

The redesigned system minimizes cervical and shoulder strain by redistributing weight, allowing workers to carry heavy loads more comfortably and reducing the risk of long-term injury.

WORKFLOW EFFICIENCY

By maintaining compatibility with traditional workflow, the new design allows for seamless integration into existing practices, enhancing productivity without disrupting established routines among workers.

ENHANCED COMFORT

Improved ergonomic features ensure greater comfort for workers during long hours, promoting better posture and reducing fatigue while harvesting tea leaves in the plantations.

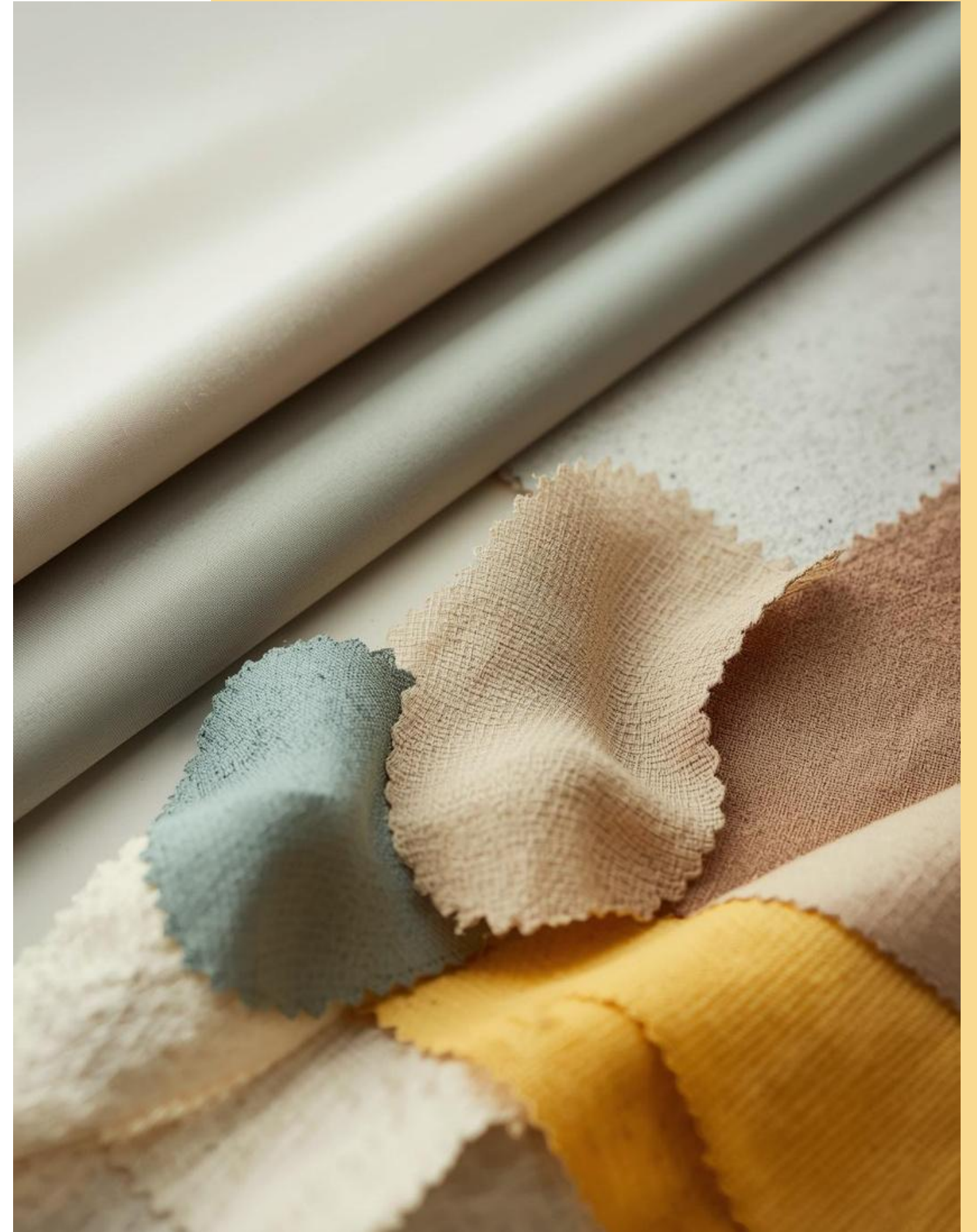
COST-EFFECTIVENESS

The affordable nature of the redesign encourages wider adoption among tea workers, promoting better safety and comfort without imposing significant financial burdens on individuals or communities.

Implementation Considerations

KEY MATERIALS AND MANUFACTURING PROCESSES

Selecting **lightweight, durable materials** is crucial. Techniques like injection molding and sustainable sourcing can enhance cost-effectiveness while ensuring **long-lasting performance** in diverse plantation conditions.



**THANK
YOU**